



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO METTUR TALUK LORRY OWNERS ASSOC

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

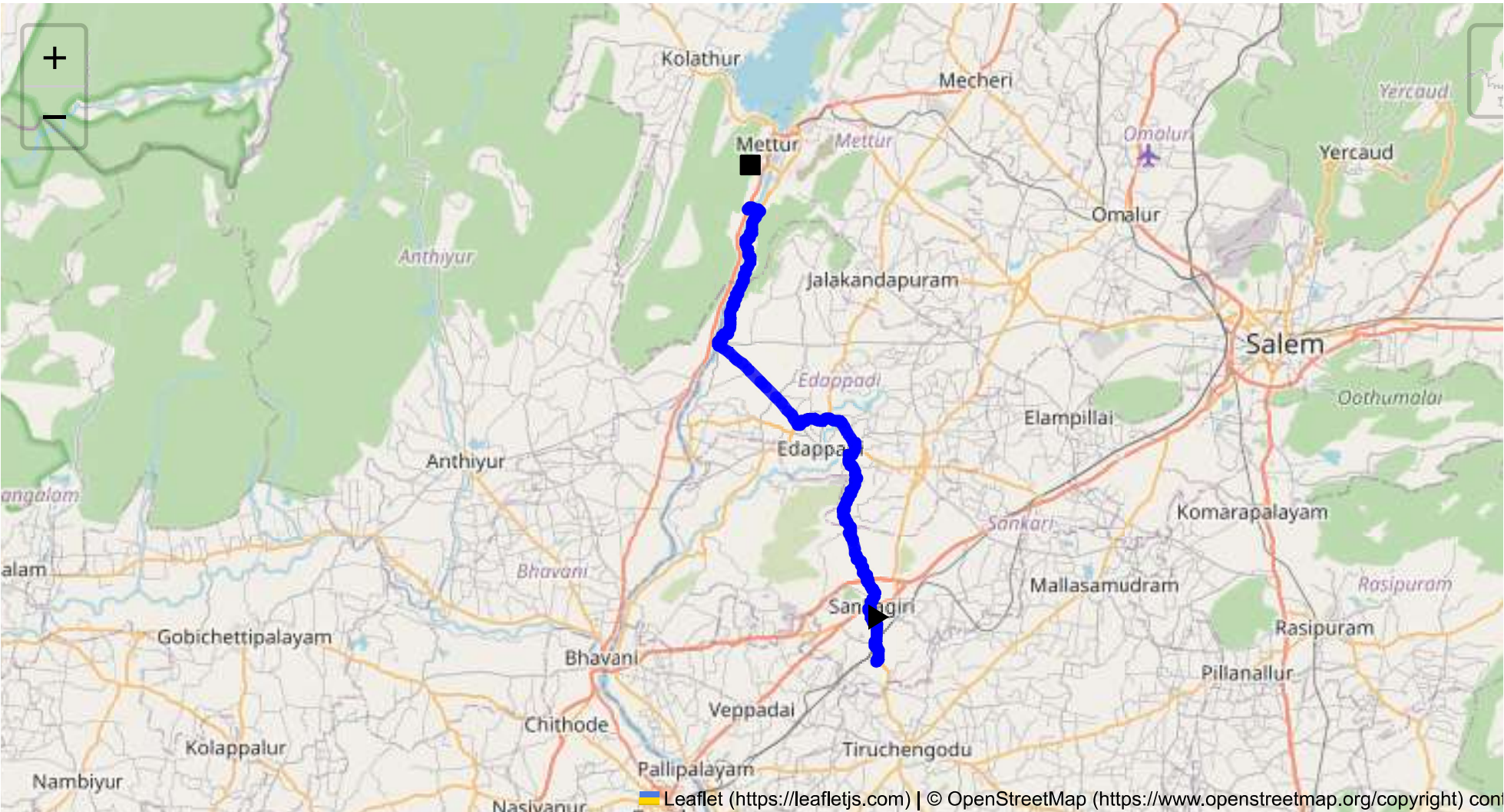
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.84 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.746061, 77.785233)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Navappatti, Tamil Nadu

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to PQWP+Q5, Navappatti spans approximately 44.84 kilometers. The journey typically includes a mix of state highways and local roads, traversing rural and semi-urban areas.

Key waypoints may include Erode-Sankari Main Road (SH-86) and connecting roads that lead towards Navappatti.

2. Typical Weather Conditions

- **Weather:** Tamil Nadu generally has a tropical climate with hot summers, moderate monsoons, and mild winters.
- **Hazards:** The region is prone to heavy rain during the monsoon season (June to September), which could cause flooding and poor visibility. Summer months can witness high temperatures, causing potential tire blowouts or dehydration risks for drivers.

3. Traffic Patterns

- **Peak Hours:** Traffic congestion typically occurs during the morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) commuter periods.
- **Congestion Areas:** Urban stretches near town centers and market areas could pose delays, especially during peak hours.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The state highways in Tamil Nadu are generally well-maintained, but local roads may experience potholes and require caution.
- **Infrastructure:** Adequate signage and lighting can be inconsistent on smaller, rural roads.

5. Alternative Routes for Emergencies

Consider routes via more populated areas or highways like AH47 for better road conditions and emergency services access. Using tech navigation can provide real-time data on alternative path availability.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu has stringent regulations on the transportation of hazardous materials, mandating permits, and predetermined routes. Compliance with these regulations is crucial, and vehicles must be appropriately marked and equipped as per guidelines.

7. Historical Incidents Involving Heavy Vehicles

The region has seen occasional accidents involving heavy vehicles, largely attributed to road conditions, driver fatigue, or inadequate vehicle maintenance. Investigate specific incident data from local agencies for comprehensive insights.

8. Environmental Considerations

- **Sensitive Areas:** Some sections might pass through agricultural zones; caution is necessary to prevent accidental spills.

- **Wildlife:** No significant wildlife areas reported, but be mindful of stray livestock.

9. Communication Coverage

Most of the route will likely have robust mobile network coverage. However, rural stretches may experience signal dead zones sporadically. It is advisable to carry reliable communication devices and backup options.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency services can typically respond within 15-30 minutes.
- **Rural Areas:** Response times can increase to 30-60 minutes due to distance and road access issues. Establish contacts with local emergency departments.

11. Overall Summary of Risk Assessment

- **Weather Risks:** High during monsoon, manageable otherwise.
- **Traffic and Road Quality:** Moderate risk with potential bottlenecks in urban areas and variability in rural road conditions.
- **Emergency Preparedness:** Necessary to enhance communication readiness and familiarize with regulatory compliance.
- **Driver Preparedness:** Necessary to train in defensive driving techniques and vehicle safety checks.

Overall, the route from Sangagiri to Navappatti presents manageable risks provided there is adherence to regulations, awareness of weather conditions, and preparedness for emergency situations. Regular updates and awareness programs for drivers on best practices can help mitigate risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	High	11.45004, 77.87206	15 KM/Hr
7	Turn	Medium	11.45028, 77.87202	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.47451, 77.86914	15 KM/Hr
11	Turn	High	11.47920, 77.86989	15 KM/Hr
12	Turn	Medium	11.48070, 77.87166	30 KM/Hr
13	Turn	Medium	11.48155, 77.87174	30 KM/Hr
14	Turn	Medium	11.48277, 77.87250	30 KM/Hr
15	Turn	Medium	11.52571, 77.85522	30 KM/Hr
16	Turn	Medium	11.53598, 77.85142	30 KM/Hr
17	Turn	Medium	11.53857, 77.85087	30 KM/Hr
18	Turn	Medium	11.55027, 77.85506	30 KM/Hr
19	Turn	Medium	11.55460, 77.85624	30 KM/Hr
20	Turn	High	11.57893, 77.85430	15 KM/Hr
21	Turn	Medium	11.57903, 77.85441	30 KM/Hr
22	Turn	Medium	11.65390, 77.76456	30 KM/Hr
23	Turn	Medium	11.65412, 77.76440	30 KM/Hr
24	Turn	High	11.65605, 77.76467	15 KM/Hr
25	Turn	High	11.65620, 77.76451	15 KM/Hr
26	Turn	Medium	11.65753, 77.76500	30 KM/Hr
27	Turn	High	11.65764, 77.76613	15 KM/Hr
28	Turn	High	11.66101, 77.76754	15 KM/Hr
29	Turn	High	11.66088, 77.76846	15 KM/Hr
30	Turn	Medium	11.66119, 77.76875	30 KM/Hr
31	Turn	Medium	11.66121, 77.76989	30 KM/Hr
32	Turn	Medium	11.66131, 77.77003	30 KM/Hr
33	Turn	High	11.66385, 77.77081	15 KM/Hr
34	Turn	High	11.66394, 77.77133	15 KM/Hr
35	Turn	Medium	11.66852, 77.77232	30 KM/Hr
36	Turn	High	11.66865, 77.77225	15 KM/Hr
37	Turn	Medium	11.66871, 77.77197	30 KM/Hr
38	Turn	High	11.66886, 77.77179	15 KM/Hr
39	Turn	Medium	11.66909, 77.77182	30 KM/Hr
40	Turn	Medium	11.66940, 77.77208	30 KM/Hr
41	Turn	Medium	11.72637, 77.78419	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	High	11.72739, 77.78410	15 KM/Hr
43	Turn	High	11.72758, 77.78426	15 KM/Hr
44	Turn	High	11.72745, 77.78539	15 KM/Hr
45	Turn	Medium	11.72777, 77.78576	30 KM/Hr
46	Turn	Medium	11.73965, 77.78829	30 KM/Hr
47	Turn	Medium	11.73995, 77.78933	30 KM/Hr
48	Blind Spot	Blind Spot	11.74531, 77.79226	10 KM/Hr
49	Turn	High	11.74807, 77.78570	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60019, 77.81968



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47451, 77.86914



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47920, 77.86989



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48070, 77.87166



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48155, 77.87174



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53598, 77.85142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53857, 77.85087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55027, 77.85506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65390, 77.76456



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65412, 77.76440



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65605, 77.76467



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65620, 77.76451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65753, 77.76500



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65764, 77.76613



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66101, 77.76754



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66088, 77.76846



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66119, 77.76875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66121, 77.76989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66131, 77.77003



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66385, 77.77081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66394, 77.77133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66852, 77.77232



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66865, 77.77225



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66871, 77.77197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66886, 77.77179



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66909, 77.77182



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66940, 77.77208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72637, 77.78419



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72739, 77.78410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72758, 77.78426



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72745, 77.78539



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72777, 77.78576



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73965, 77.78829



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73995, 77.78933



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.74531, 77.79226



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74807, 77.78570

